Mohammad Vazir Panah

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Nationality: Iranian · DOB: 23 Apr 1994



Summary

High-performance and embedded computing researcher working on compiler and runtime techniques for performance-portable SYCL programming across heterogeneous RISC-V architectures, from embedded CPUs to GPUs. Experience with auto-vectorization and code generation for RISC-V vector processors (GCC, LLVM), parallel programming (SYCL, OpenMP, OpenCL, CUDA) and performance analysis on real hardware. Background in FPGA and digital hardware design, real-time embedded systems and formal modeling of cyber-physical systems (Robotics domain).

Research Interests

High-Performance Computing, Compilers & Auto-Vectorization, Performance Portability & Heterogeneous Computing, RISC-V Architectures, Parallel Programming Models, Real-Time & Embedded Systems, Formal Modeling & Verification

Education

M.Sc. in Computer Engineering, Computer Systems Architecture *Sep 2019 - Oct 2022*
Shahid Beheshti University, Tehran, Iran

- GPA: 18.43 / 20.00
- Thesis: *A Framework for Modeling Self-Aware Cyber-Physical Systems with SystemC*
- Supervisor: Dr. Seyed-Hosein Attarzadeh-Niaki
- Advisor: Dr. Armin Salimi-Badr

B.Sc. in Computer Engineering, Hardware Engineering *Sep 2012 - May 2016*
Birjand University, Birjand, Iran

- GPA: 16.00 / 20.00
- Project: *Simulation and Programming the Alpha Rex Human-Robot Produced by Lego Company*

Work Experience

Research Fellow (Assegno di Ricerca) *2023 - 2026*
***University of Salerno**, Dept. of Computer Science, ISISLab, Salerno, Italy*

- Scientific Responsible: Prof. Biagio Cosenza
- Research bridging real-time embedded systems and high-performance heterogeneous computing across the RISC-V computing continuum, from embedded cores to HPC vector processors
- Benchmarked GCC, LLVM, and LLVM-EPI auto-vectorization for the RISC-V Vector Extension (RVV 0.7 & 1.0) across 151 TSVC loops and seven real-world applications on two RISC-V vector boards; showed LMUL tuning yields up to 3× speedup and characterized per-pattern compiler limitations. Published at IEEE PDP 2025 (2nd author)
- Contributed to LibreRT: developing and deploying portable real-time applications across heterogeneous embedded CPU/GPU platforms (SYCL, OpenMP, OpenCL)
- Fixed SYCL/AdaptiveCPP kernel dispatch failures on Imagination PowerVR embedded GPUs by tracing them into the LLVM-based runtime
- EuroHPC LIGATE project: set up RISC-V toolchains, firmware and embedded software stacks on physical boards shared across partner institutions
- Built reproducible experiment pipelines (Git, Docker, CI, CMake) for RISC-V/SoC boards; documented setups for reproducibility

Instructor *2013 - 2015*
***Sepehr Educational Institutions**, Kashmar, Iran*

- Delivered theoretical and practical instruction in C++ programming
- Taught ICDL (International Computer Driving License) skills

- Taught CorelDRAW and Adobe FreeHand, focusing on vector graphics

Education Intern *2014 - 2015*
***Abooreyhan Company**, Kashmar, Iran*

- Hardware/software troubleshooting, diagnostics, and component replacement
- Assembled and configured PCs; software installation and system setup

Publications

2026 **LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum**
*B. Cosenza, F. Terraneo, G. Agosta, L. Carpentieri, D. Cattaneo, **M. Vazir Panah**, N. Nicolosi*
23rd ACM International Conference on Computing Frontiers (CF), Special Sessions · 2026

2025 **A Performance Analysis of Autovectorization on RVV RISC-V Boards**
*L. Carpentieri, **M. Vazir Panah**, B. Cosenza*
PDP, 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing, IEEE · 2025

2023 **System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs**
*S.-H. Attarzadeh-Niaki, **M. Vazir Panah***
Journal of Innovations in Computer Science and Engineering (JICSE), Shahid Beheshti University · 2023

2022 **ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design**
***M. Vazir Panah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr*
27th International Computer Conference, Computer Society of Iran (CSICC), IEEE · 2022

In Prep. **Formal Modeling of Self-Aware CPS via SADF and ROS/Gazebo Co-Simulation**
***M. Vazir Panah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr*
In preparation · 2026

Academic Experience

Teaching Assistantships

Design & Analysis of Real-Time Embedded Systems	Shahid Beheshti University	2021
System-on-Chip	Shahid Beheshti University	2020, 2021
Advanced Computer Architecture	Shahid Beheshti University	2021
Fundamentals of Embedded & Real-Time Systems	Shahid Beheshti University	2020
Computer Architecture	Birjand University	2016

Laboratory Memberships

- ISISLab, HPC Group, University of Salerno, Italy
- SBU Cyber-Physical Systems Lab Research Team, Shahid Beheshti University, Tehran, Iran

Selected Projects

RISC-V Benchmark Suite *2025*
Open-source project GitHub

- Designed and implemented an open-source benchmark suite for RISC-V and RVV architectures, covering diverse data-level-parallelism patterns
- Evaluated auto-vectorization performance across multiple GCC and LLVM compiler versions to isolate cost factors

LibreRT, Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum *2023 - 2026*
Researcher, Italian PRIN 2022-funded project

- Developed and evaluated real-time applications for heterogeneous embedded CPU/GPU platforms
- Implemented solutions using SYCL, OpenMP, OpenCL, and RISC-V boards

Parallel Sorting & Matrix Multiplication Architectures *2020*
Course Project, M.Sc. Parallel Systems Architecture, Open-source project GitHub

- Designed mesh- and hypercube-connected sorting-network processors in Verilog, with a CUDA implementation of the hypercube variant
- Designed a multistage-interconnection-network-based shared-memory matrix multiplication architecture in Verilog

Self-Aware CPS Modeling Framework *2022*

M.Sc. Thesis, Open-source project GitHub, based on ForSyDe-SystemC Modeling Framework

- Designed and implemented a self-aware cyber-physical system model using ForSyDe-SystemC and ROS-Gazebo co-simulation
- Developed an SADF-based model and contributed extensions to the ForSyDe-SystemC library; the core SADF process constructors were merged into the official ForSyDe-SystemC repository

FIR Filter and Tiny YOLO v1 on FPGA *2019*

Course Project, M.Sc. System-on-Chip, DE0-Nano / DE10 Intel FPGA Board, Open-source project GitHub

- Designed and implemented a FIR filter IP core in Quartus with an Avalon-MM interface
- Deployed a real-time Tiny YOLO v1 object-detection (CNN) inference pipeline on FPGA under tight resource constraints; an edge-AI / ML inference workload on constrained hardware

Full-Scan DFT, BIST & LSSD Self-Test for a Custom CPU *2020*

Course Project, M.Sc. Test and Testability, Open-source project GitHub

- Applied full-scan design-for-test methodology (netlist unfolding, fault collapsing, ATPG, scan-chain insertion) to a custom-designed CPU, reaching 100% fault coverage after scan insertion
- Designed and evaluated BIST (LFSR/MISR-based Random Test Socket) and LSSD on-chip self-test (boundary-scan cells + on-chip signature monitor) architectures for the same design

Real-Time Audio Controller, UPPAAL Model Checking & Lingua-Franca Implementation *2020*

Course Project, M.Sc. Design and Analysis of Real-Time Embedded Systems, Open-source project GitHub

- Modeled a real-time audio-controller case study as timed automata in UPPAAL and verified timing-correctness properties
- Re-implemented the model as a reactive program in Lingua-Franca, resolving clock-drift between distributed timers

Railroad System Verification with NuSMV *2020*

Course Project, M.Sc. Design and Analysis of Real-Time Embedded Systems, Open-source project GitHub

- Modeled and verified a railroad control system using NuSMV
- Checked safety and liveness properties through formal model checking

Technical Skills

Programming	C, C++, Python, Assembly
Heterogeneous & HPC	SYCL (AdaptiveCPP), OpenMP (including SIMD), OpenCL, CUDA, SIMD intrinsics
Compilers & Toolchains	GCC, LLVM/Clang, auto-vectorization analysis, RISC-V Vector extension (RVV 0.7 & 1.0), cross-compilation
Methods & Concepts	Parallel & Vector Computing, Performance Portability, Real-time & Embedded systems, Formal Verification & Model Checking
Hardware & Embedded Systems	RISC-V, FPGA, VHDL, Verilog, FreeRTOS, high-level synthesis (basic)
Simulation & EDA	Gem5, QEMU, OVPsim, Quartus, ModelSim, MATLAB-Simulink
Modeling & Verification	SystemC, SystemC-AMS, NuSMV, UPPAAL
Robotics	ROS, Gazebo, Sensor Integration
Tools	Git, Docker, Linux, CMake, LaTeX
Certifications	TinyML, Harvard University, edX

Honours and Awards

National R&D Research Grant Winner *2020 - 2021*

R&D Center, Mobile Telecommunication Company of Iran (MCI)

- Awarded among competing Master's and PhD students in a national grant competition

Top Computer Engineering Student *2016*
Birjand University

- Ranked in the top 15% of B.Sc. Computer Engineering graduates

Conference Participation

33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP) *Mar 2025*
Co-author and attendee, Turin, Italy

- Co-author of the paper: *A Performance Analysis of Autovectorization on RVV RISC-V Boards*

27th International Computer Conference, Computer Society of Iran (CSICC) *Feb 2022*
Presenter, Sharif University of Technology, Tehran, Iran

- Presented the paper: *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design*

Language Skills

	Listening	Reading	Writing	Speaking
Persian	Native	Native	Native	Native
English	C1	C1	C1	C1
Italian	B1	A2	A2	A2

References

Prof. Biagio Cosenza
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University of Salerno, Italy
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